

Lesson plan

Part A. Course: Astronomy 101, an introductory astronomy course for STEM and non STEM majors. This course would likely have a range of students, freshmen to seniors and would likely be pretty large, maybe about 100 students or so. The class would be in a lecture hall twice a week for 50 minutes. In this course we would cover topics such as, telescopes (why we use them, how they work), moon phases, orbits, parallax (measuring distances in space), distance ladder and standard candles, exoplanet detection, the Milky Way, and rotation curves and the mass of the galaxy. For this lesson I will focus on **moon phases**.

Part B. Learning Objectives: Moon Phases

- Students will be able to describe the moon phases and tell the difference between waxing and waning
- Students will be able to explain why we see moon phases and the time it takes to complete one cycle
- Students will be able to apply this knowledge to different scenarios.

Part C. Activities:

Students will be in groups and have a globe, a flashlight, and a small ball (the moon). First students will be instructed to show how the Earth orbits around the sun, no moon. One student will be the sun and will still in one place, while another student orbits their Earth around them. They will discuss their observations and then add in the moon. Students will play out what they think happens and then describe what they see. They are encouraged to try out different scenarios.

Part D. Assessments

During the class I will use clicker questions for informal assessment.

First I will give a short lecture where I show the different moon phases and explain the difference between waxing and waning. I will then ask the students to describe their observations of the moon's phases. This phase of the class will be where students volunteer information we can use to better understand the question, why does the moon have phases? Then I will ask the class where is the light coming from? - answer I'm looking for is, the light is from the sun and is reflected off the surface of the moon.

If students struggle to get here then lets assume that the moon emits its own light, what would we see then? (Constant full moon, no phases)

Recap: Yes when the moon is dark the other half is being lit up by the sun. When the moon is full the half we see is the half illuminated by the sun.

Then I will pose the question why does the moon have phases with the following responses.

- A. The moon is orbiting around the Earth and we see different portions of it hit by sunlight
- B. The moon is fixed in one point near the Earth but is rotating so only portions of it are hit by sunlight.
- C. The moon dies and is reborn again, the phases are the stages of its life

After this questions students will pair up with two others and will be given the activity. After the activity they will be asked this same question again.

This time though I will ask for people to explain their reasoning.

The final assessment on this lesson will be a homework problem describing a different planetary scenario. This will take the concepts learned in class and apply it to a different situation. Students are encouraged to work together so long as they credit each other. I may ask a similar question on a test.

Questions about your lesson plan

Objectives

How have you followed recommended practices in identifying your learning objectives? In particular, have you expressed your goals in terms of what students will achieve or be able to do? Are your goals well-defined and measurable? Are they at appropriate levels of Bloom's taxonomy?

Yes, I wrote all learning objectives as "students will be able to.." Statements. I started with a lower level Bloom's word "describe", then "explain" and then "apply". So the learning objectives get more advanced but start small.

Misconceptions

What potential student misconceptions have you identified in your lesson plan? How have you planned to surface and respond to these misconceptions?

Where does the light come from? Why do we see different illuminated portions? (orbit vs rotating) The first misconception is hopefully not hard to dispel. For this one I just posed the question to the class and had them explain using prior knowledge. For the second one we use an activity where students use a flashlight as the sun and either rotate their moon or orbit it around the Earth to see what better matches what we see.

Practice & Feedback

What opportunities for students to have meaningful practice and feedback have you planned for this lesson? What kinds of feedback would you expect students to receive and from whom?

Students will be able to get feedback from clicker questions and work in groups to get more feedback from peers. The activity will give them practice with both ideas to see which makes more sense.

Sequencing

Why have you sequenced the learning activities you describe in your lesson plan in the order you have planned them?

I wanted to break up the concept in two since there are two main components, where is the light coming from and why do we see the phases.

If we first understand where the light is coming from it makes it easier to understand why we see the phases.

I do want to ask the students where the phases come from before doing the activity so that they can test their theory using an experiment. (Gets them to think like scientists and also be ok with being “wrong” because its intentionally apart of the learning process).

Last is final assessment and recap to really nail down the concept so they can go home and work on the homework problem.

Alignment

What are 2 or 3 ways your objectives, assessments, and activities are in alignment? Be specific about connections you have planned among these three components.

The learning objectives use the words, describe, explain and apply. The activity asks students to describe what they see and explain how it is working. The assessments ask the students to apply this knowledge to answer questions.

We start by asking students to describe their own experience, to get them thinking about what they see. This utilizes prior knowledge but they might need more help being able to explain it. That is the next step, where hopefully the activity helps them see the mechanics behind what they see. The application asks students to use this new knowledge with their prior knowledge and think about different scenarios. What if we tweaked the mechanics a bit, what if they saw something a little different, would they be able to explain why?

Evidence-Based Practices

How has research on STEM teaching and learning, including but not limited to studies mentioned in this course, informed the teaching choices you made as you planned your lesson?

The studies on peer based learning really informed this lesson. I think it lowers the stakes a bit, they aren't just accepting information from me but are building it with their peers. This also

makes a more comfortable environment where students can more freely make mistakes. And by working closely with their peers regularly my hope is that they will build bonds with their classmates.

Inclusive Teaching

What aspects of your lesson plan are intended to create a learning environment that welcomes full participation by all students? Have you planned activities that might leverage the diversity of perspectives and experiences among your students?

This lesson begins by students talking about their experience and using that experience as key scientific information. All students are encouraged to participate but I understand not every student feels comfortable sharing with the whole class. Breaking them into small groups helps with this and then also having the homework at the end where they can work alone or in groups.